$\sqrt{7}$ is Irrational

Proof that $\sqrt{7}$ is irrational

Permitted external facts

We use the following standard facts without further proof.

1. **The Division Algorithm:** for any integers a and d with $d > 0$, there exist unique integers q and r with

$$a = dq + r \quad \text{and} \quad 0 \leq r < d.$$

   By uniqueness, $r = 0$ if and only if $d \mid a$, and $r \neq 0$ if and only if $d \nmid a$.

2. **Euclid's Lemma:** if p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

3. **The Well-Ordering Principle:** every nonempty set of positive integers has a least element.

4. **Closure:** the integers are closed under addition, subtraction, and multiplication.

5. **Divisibility definition:** $d \mid a$ means there exists an integer k such that $a = dk$. In particular, $1 \mid n$ for every integer n (since $n = 1 \cdot n$), so 1 is always a divisor.

6. **Ordered-ring and ordered-field properties (freely used throughout):**

   - *Trichotomy:* every nonzero integer (or real number) is either positive or negative; in particular, if $b < 0$ then $-b > 0$.
   - *Integer discreteness:* every positive integer n satisfies $n \geq 1$.
   - *Product of positives:* if $a > 0$ and $b > 0$ (integers, rationals, or reals), then $ab > 0$.
   - *Quotient of positives:* if $a > 0$ and $b > 0$ then $a/b > 0$.
   - *Order compatibility with multiplication:* if $a \geq b$ and $c > 0$, then $ac \geq bc$; if $a > b$ and $c > 0$, then $ac > bc$.
   - *Order compatibility with addition:* $x > y$ if and only if $x - y > 0$.

- *Integer discreteness (lower bound):* if n is an integer with $n > k$ for some integer k, then $n \geq k+1$. In particular, $n > 0$ implies $n \geq 1$, and $n > 1$ implies $n \geq 2$.
- *Cancellation/division:* if $a = bc$ and $c \neq 0$, then $a/c = b$.
- *Fraction cancellation:* if $c \neq 0$, then $(ac)/(bc) = a/b$.
- *Negation preserves equality:* $(-a)/(-b) = a/b$ for $b \neq 0$.
- *Squaring a fraction:* $(a/b)^2 = a^2/b^2$ for $b \neq 0$.
- *Positive square root:* $\sqrt{x}$ denotes the unique positive real satisfying $(\sqrt{x})^2 = x$, for $x > 0$.

Lemma 1. *For any integer n, if $7 \mid n^2$ then $7 \mid n$.*

Proof. **Step 1. Primality of 7.**

A positive integer p is *prime* if $p > 1$ and its only positive divisors are 1 and p. Since $7 = 6 + 1 > 1$, the condition $7 > 1$ is satisfied. We verify that no integer d with $2 \leq d \leq 6$ divides 7, using the Division Algorithm (in each case the remainder r satisfies $0 \leq r < d$, confirmed by inspection):

$$7 = 2 \cdot 3 + 1, \quad r_2 = 1,\ 0 \leq 1 < 2, \quad r_2 \neq 0, \quad \text{so } 2 \nmid 7;$$

$$7 = 3 \cdot 2 + 1, \quad r_3 = 1,\ 0 \leq 1 < 3, \quad r_3 \neq 0, \quad \text{so } 3 \nmid 7;$$

$$7 = 4 \cdot 1 + 3, \quad r_4 = 3,\ 0 \leq 3 < 4, \quad r_4 \neq 0, \quad \text{so } 4 \nmid 7;$$

$$7 = 5 \cdot 1 + 2, \quad r_5 = 2,\ 0 \leq 2 < 5, \quad r_5 \neq 0, \quad \text{so } 5 \nmid 7;$$

$$7 = 6 \cdot 1 + 1, \quad r_6 = 1,\ 0 \leq 1 < 6, \quad r_6 \neq 0, \quad \text{so } 6 \nmid 7.$$

We now argue the case split is exhaustive. Let $d > 1$ be any positive divisor of 7; by the divisibility definition, $7 = dk$ for some integer k. Since $7 > 0$ and $d > 0$, by the quotient-of-positives property $k = 7/d > 0$, so k is a positive integer. By integer discreteness, $k \geq 1$. Therefore $d = 7/k \leq 7/1 = 7$ (by the cancellation property and order compatibility with multiplication: $k \geq 1$ and $d > 0$ imply $dk \geq d \cdot 1 = d$, giving $d \leq dk = 7$). Thus every positive divisor $d > 1$ of 7 satisfies $2 \leq d \leq 7$. For $2 \leq d \leq 6$ we checked above that $d \nmid 7$. For $d = 1$: $7 = 1 \cdot 7$ and $7 \in \mathbb{Z}$, so $1 \mid 7$ by the divisibility definition. For $d = 7$: $7 = 7 \cdot 1$ and $1 \in \mathbb{Z}$, so $7 \mid 7$ by the divisibility definition. Every positive divisor $d > 1$ of 7 satisfies $2 \leq d \leq 7$: the lower bound $d \geq 2$ follows from integer discreteness applied to $d > 1$ (since d is a positive integer with $d > 1$, integer discreteness gives $d \geq 2$); the upper bound $d \leq 7$ was established above. Since d is an integer with $2 \leq d \leq 7$, it belongs to $\{2, 3, 4, 5, 6, 7\}$, and we have checked all cases. Therefore the only positive divisors of 7 are 1 and 7, confirming that 7 is prime.

Step 2. Apply Euclid's Lemma.

Suppose $7 \mid n^2$. Since the integers are closed under multiplication, $n^2 = n \cdot n$ is an integer. Since 7 is prime (Step 1) and $7 \mid n \cdot n$, Euclid's Lemma gives $7 \mid n$ or $7 \mid n$; in either case $7 \mid n$. □

Theorem 1. *$\sqrt{7}$ is irrational.*

Proof. **Step 1. Setup.**

Assume for contradiction that $\sqrt{7}$ is rational. By definition there exist integers a, b with $b \neq 0$ such that $\sqrt{7} = a/b$. By trichotomy, either $b > 0$ or $b < 0$ (since $b \neq 0$). If $b < 0$, replace (a, b) by $(-a, -b)$: since $b < 0$, we have $-b > 0$ by trichotomy, and $(-a)/(-b) = a/b$ by the negation-preserves-equality property, so the equation $\sqrt{7} = a/b$ is preserved. In either case, we may assume $b > 0$. Since $\sqrt{7}$ is the positive square root of 7 and $7 > 0$, we have $\sqrt{7} > 0$. Multiplying both sides of $\sqrt{7} = a/b$ by $b > 0$ gives $a = b\sqrt{7}$. Since $b > 0$ and $\sqrt{7} > 0$, the product $a = b\sqrt{7}$ is a product of two positive numbers, hence $a > 0$.

Step 2. Well-ordering.

Define

$$S = \{q \in \mathbb{Z}_{>0} : q\sqrt{7} \in \mathbb{Z}_{>0}\}.$$

Since $b \in \mathbb{Z}_{>0}$ and $b\sqrt{7} = a \in \mathbb{Z}_{>0}$, we have $b \in S$, so S is nonempty. By the Well-Ordering Principle, S has a least element q_0. Set $p_0 = q_0\sqrt{7} \in \mathbb{Z}_{>0}$. Then $p_0 > 0$ (since $q_0 > 0$ and $\sqrt{7} > 0$) and $\sqrt{7} = p_0/q_0$.

Step 3. 7 divides p_0.

Squaring both sides of $\sqrt{7} = p_0/q_0$ gives $(\sqrt{7})^2 = (p_0/q_0)^2$, i.e., $7 = p_0^2/q_0^2$. Since $q_0 \in \mathbb{Z}_{>0}$, the square $q_0^2 = q_0 \cdot q_0$ is a positive integer by closure. Multiplying both sides by q_0^2 yields $p_0^2 = 7q_0^2$. Hence $7 \mid p_0^2$. By Lemma 1 (with $n = p_0 \in \mathbb{Z}$), $7 \mid p_0$. By the divisibility definition, there exists an integer k such that $p_0 = 7k$. Since $p_0 > 0$ and $7 > 0$, we have $k = p_0/7 > 0$, so $k \in \mathbb{Z}_{>0}$.

Step 4. 7 divides q_0.

Substituting $p_0 = 7k$ into $p_0^2 = 7q_0^2$ gives $(7k)^2 = 7q_0^2$, i.e., $49k^2 = 7q_0^2$. Since $7 \neq 0$, dividing both sides by 7 gives $7k^2 = q_0^2$. Since $k > 0$, we have $k^2 = k \cdot k > 0$, and k^2 is an integer by closure. Hence $7 \mid q_0^2$. By Lemma 1 (with $n = q_0 \in \mathbb{Z}$), $7 \mid q_0$. By the divisibility definition, there exists an integer m such that $q_0 = 7m$. Since $q_0 > 0$ and $7 > 0$, we have $m = q_0/7 > 0$, so $m \in \mathbb{Z}_{>0}$.

Step 5. Contradiction.

From $\sqrt{7} = p_0/q_0 = 7k/(7m)$, since $m > 0$ (hence $m \neq 0$) and $7 \neq 0$, we may cancel the common factor 7 in numerator and denominator (by fraction cancellation) to obtain $\sqrt{7} = k/m$. Multiplying both sides by $m > 0$ gives $m\sqrt{7} = k \in \mathbb{Z}_{>0}$, so $m \in S$.

It remains to show $m < q_0$, contradicting the minimality of q_0. Since $q_0 = 7m$ and $7 \geq 2$ (by inspection: $7 = 2+5$), the order-compatibility-with-multiplication property gives $7m \geq 2m$ (multiplying $7 \geq 2$ by $m > 0$). Also, $2m - m = m > 0$, and by order-compatibility-with-addition ($x - y > 0 \Leftrightarrow x > y$), this gives $2m > m$. Therefore $q_0 = 7m \geq 2m > m$, i.e., $m < q_0$. Thus $m \in S$ and $m < q_0$, contradicting the minimality of q_0.

Therefore the assumption that $\sqrt{7}$ is rational is false, and $\sqrt{7}$ is irrational. $\square$